



ECO 7115 | Microeconomic Theory 1

Class Number: 11168

Fall 2026

Instructor Information

Name: Fatma Gunay

Teaching Assistants

Name: Chenhui Lu

Email: ch.lu@ufl.edu

Office Hours: TBA

Class Number: ECO7115

Course Details

Catalog Description: Analysis of optimization applied to consumer and product theory including comparative statistics and duality.

Credit Hours: 3

Course Fees: \$0.00

Additional Course Description

This doctoral-level course comprises the initial segment of a sequential exploration into the underpinnings of contemporary microeconomic theory. The first part of this course delves into decision theory, expounding upon the ramifications of rational decision-making within diverse contexts. Subsequently, the second part delves into the equilibrium interplay among rational decision makers, encompassing partial equilibrium theory and lotteries.

The course presupposes that students have completed an intermediate microeconomics course at the undergraduate level. Additionally, it anticipates a solid grasp of multivariable calculus, linear algebra, and fundamental real analysis on the part of the students.

Required Materials

MICROECONOMIC THEORY

ISBN: 0195073401

Authors: MAS-COLELL, WHINSTON & GREENE

Publisher: OXFORD UNIVERSITY PRESS

Edition: 1

Recommended Materials

Advanced Microeconomic Theory

Authors: Jehle, Geoffrey A, Reny, Philip J.

Fundamental Methods of Mathematical Economics

Authors: Chiang, A.C., Wainwright, K.

Recommended Text for Math Review

Microeconomic Analysis

Authors: Varian, Hal R

Understanding Analysis

Authors: Abbott, S.

Recommended Text for Basic Real Analysis

A Note on Materials

The textbook is required for the course.

Course Goals and Objectives

The objective of this course is to provide students with the theoretical and analytical foundation necessary for advanced doctoral study and research in economics. Upon successful completion of the course, students should be able to formulate and solve formal microeconomic models, apply optimization techniques to analyze economic behavior, evaluate decisions under uncertainty, and use the tools of modern microeconomic theory to study markets, firms, and economic welfare.

Expectations and Student Learning Outcomes

This course provides a rigorous introduction to modern microeconomic theory at the doctoral level. The primary goal is to develop the analytical tools necessary for advanced study and research in economics. By the end of the course, students should be able to:

1. Develop and analyze consumer theory models, including preferences, utility maximization, expenditure minimization, and duality.
2. Analyze individual decision-making under certainty and uncertainty using the axiomatic approach to consumer choice and expected utility theory.
3. Understand and apply theories of choice under risk, including expected utility, risk aversion, and lottery-based decision making.
4. Analyze producer behavior through profit maximization, cost minimization, production functions, and duality theory.
5. Evaluate competitive market outcomes using partial equilibrium analysis and welfare concepts.
6. Build a foundation for advanced doctoral coursework in game theory, general equilibrium, information economics, mechanism design, and other fields of microeconomic theory.

Methods of Evaluation

GRADING SCHEME		
75%	Exams	25%, each
25%	HW Assignments	2.5%, each

Grading Scale

The grading scale presented below is tentative and may be adjusted at the instructor's discretion. Any modifications to the grading scale will be made only if they benefit students and will never result in a lower course grade than would have been earned under the original scale. Any changes will be communicated to students in a timely manner.

Letter grade and percentage

Letter	Percentage Value
A	94 - 100%
A -	90 - 93%
B +	87 - 89%
B	83 - 86%
B -	80 - 82%
C +	77 - 79%
C	73 - 76%
C -	70 - 72%
D +	67 - 69%
D	63 - 66%
D -	60 - 62%
E	59% and below

Course Schedule

Course Schedule				
Week	Dates	Class Dates	Topics	Readings (required and supplemental)
Week 1	Aug 20-23	Aug 20	Preference Relations; Revealed Preference; Utility Representation; Continuous Utility	MWG Ch. 1; JR Ch. 1; Varian Ch. 7
Week 2	Aug 24-30	Aug 25, Aug 27	Consumer Choice; Utility Maximization	MWG Ch. 3; JR Ch. 1; Varian Ch. 8-9
Week 3	Aug 31-Sept 6	Sept 1, Sept 3	Marshallian Demand; Comparative Statics	MWG Ch. 3; JR Ch. 1
Week 4	Sept 7-13	Sept 8, Sept 10	Expenditure Minimization; Hicksian Demand	MWG Ch. 3; JR Ch. 2
Week 5	Sept 14-20	Sept 15, Sept 17	Duality; Expenditure Functions; Slutsky Equation; Applications	MWG Ch. 3; JR Ch. 2
Week 6	Sept 21-27	Sept 22, Sept 24	Exam I and Review	
Week 7	Sept 28-Oct 4	Sept 29, Oct 1	Lotteries; Expected Utility	MWG Ch. 6; JR Ch. 2; Varian Ch. 11
Week 8	Oct 5-11	Oct 6, Oct 8	Risk Aversion; Certainty Equivalents	MWG Ch. 6
Week 9	Oct 12-18	Oct 13, Oct 15	Stochastic Dominance; Insurance Applications	MWG Ch. 6
Week 10	Oct 19-25	Oct 20, Oct 22	Exam II; Introduction to Production	MWG Ch. 5
Week 11	Oct 26-Nov 1	Oct 27, Oct 29	Production Sets; Profit Maximization	MWG Ch. 5; JR Ch. 3
Week 12	Nov 2-8	Nov 3, Nov 5	Cost Minimization; Producer Duality	MWG Ch. 5; JR Ch. 3
Week 13	Nov 9-15	Nov 10, Nov 12	Competitive Equilibrium; Pareto Efficiency	MWG Ch. 10; JR Ch. 4
Week 14	Nov 16-22	Nov 17, Nov 19	Welfare Theorems	MWG Ch. 10; JR Ch. 4
Week 15	Nov 23-29	Nov 24	Thanksgiving Break	

Course Schedule				
Week 16	Nov 30-Dec 6	Dec 1, Dec 3	Review; Exam III	MWG Ch. 10; JR Ch. 4; Varian Ch. 31

University Policies and Resources

Information about grading policies, support for students with disabilities, course evaluations, the Honor Code, and other course policies and campus resources can be found on the [Syllabus Policies page](#).

Attendance Policy

Excused and Unexcused Absences

Students may only participate in classes if they are registered officially or approved to audit with evidence of having paid audit fees. The Office of the University Registrar provides official class rolls to instructors.

Students are responsible for satisfying all academic objectives as defined by the instructor. Absences count from the first-class meeting.

Acceptable reasons for absence from or failure to engage in class include illness; Title IX-related situations; serious accidents or emergencies affecting the student, their roommates, or their family; special curricular requirements (e.g., judging trips, field trips, professional conferences); military obligation; severe weather conditions that prevent class participation; religious holidays; participation in official university activities (e.g., music performances, athletic competition, debate); and court-imposed legal obligations (e.g., jury duty or subpoena). Other reasons (e.g., a job interview or club activity) may be deemed acceptable if approved by the instructor.

For all planned absences, a student in a situation that allows an excused absence from a class, or any required class activity must inform the instructor as early as possible prior to the class. For all unplanned absences because of accidents or emergency situations, students should contact their instructor as soon as conditions permit.

Students shall be permitted a reasonable amount of time to make up the material or activities covered during absence from class or inability to engage in class activities because of the reasons outlined above.

If a student does not participate in at least one of the first two class meetings of a course or laboratory in which they are registered, and they have not contacted the department to indicate their intent, the student can be dropped from the course. Students must not assume that they will be dropped, however. The department will notify students if they have been dropped from a course or laboratory.

The university recognizes the right of the instructor to make attendance mandatory and require documentation for absences (except for religious holidays), missed work, or inability to fully engage in class. After due warning, an instructor can prohibit further attendance and subsequently assign a failing grade for excessive absences.

Religious Holidays Guidelines

At the University of Florida, students and faculty work together to allow students the opportunity to observe the holy days of their faith. A student should inform the faculty member of the religious observances of their faith that will conflict with class attendance, with tests or examinations, or with other class activities prior to the class or occurrence of that test or activity. The faculty member is then obligated to accommodate that particular student's religious observances. Because students represent a myriad of cultures and many faiths, the University of Florida is not able to assure that scheduled academic activities do not conflict with the holy days of all religious groups. Accordingly, individual students should make their need for an excused absence known in advance of the scheduled activities.

The Florida Board of Education and state law govern university policy regarding observance of religious holidays.

Guidelines

- Students, upon prior notification to their instructors, shall be excused from class or other scheduled academic activity to observe a religious holy day of their faith.
- Students shall be permitted a reasonable amount of time to make up the material or activities covered in their absence.
- Students shall not be penalized due to absence from class or other scheduled academic activity because of religious observances.

If a faculty member is informed of or is aware that a significant number of students are likely to be absent from class because of a religious observance, the faculty member should not schedule a major exam or other academic event at that time.

A student who is to be excused from class for a religious observance is not required to provide a second party certification of the reason for the absence. Furthermore, a student who believes that they have been unreasonably denied an education benefit due to religious beliefs or practices may seek redress through the student grievance procedure.

Absence due to Illness

A student who is absent from class or any required class-related activity because of illness should contact their instructor, if feasible, as early as possible prior to the missed class or activity.

Students shall be permitted a reasonable amount of time to make up the material or activities covered during an excused absence.

Students should contact their college by the deadline to drop a course for medical reasons. Students can petition the Dean of Students Office to drop a course for medical reasons. The university's policy regarding medical excuse from classes is maintained by the Student Health Care Center.

Twelve-Day Rule

Students who participate in university-sponsored athletic or scholarly activities are permitted to be absent 12 scholastic days per semester without penalty. A scholastic day is any day on which regular class work is scheduled as defined in the approved university calendar. [More Info](#)

The student or student's advisor must notify the instructor as early as possible prior to the anticipated absence to allow ample time for accommodations. Instructors must be flexible and not penalize students when re-scheduling during-term and final exams, class assignments, and other required activities and must follow the UF Attendance Policy herein and UF Examination Policies. As noted in the UF Examination Policies, during-term exams should be re-scheduled no later than

before the end of the semester, while final exams no later than 90 days after the originally scheduled exam time. However, instructors are encouraged to re-schedule final and during-term exams, assignments, and other activities as soon as possible after the last day of the absence and must not penalize the student in any way. [More Info](#)

A group's schedule that requires absence of more than 12 scholastic days should be adjusted so that no student is absent from campus more than 12 scholastic days. Students who previously have been warned in writing by their instructor about the impact of absences on their individual class performance should not incur additional absences, even if they have not been absent 12 scholastic days. The student is responsible to maintain satisfactory academic performance and attendance.

Course Policies and Resources

Exam Policies:

The course will include three in-class exams. These exams are scheduled for the following dates:

- September 22
- October 20
- December 1

It is important to note that the exams are not comprehensive; they will focus on the material covered since the previous exam.

Homework Assignments:

Throughout the duration of the course, you will be assigned weekly assignments to be completed over the weekend. These assignments will be made available every Thursday and are due for submission on the following Tuesday at 11:45am, in class.

It is imperative that you adhere to these due dates as outlined. Late submissions may be permitted at the instructor's discretion if arrangements are made with the instructor before the assignment deadline. Requests made after the deadline may not be approved.

Over the course of the semester, you will complete a total of 10 assignments. At the conclusion of the term, the lowest grade among these assignments will be dropped, offering you the opportunity to alleviate any undue pressure resulting from potential setbacks. It is highly advised that you engage in independent attempts at solving the problem sets. In the event of challenges, seeking assistance from fellow classmates, the teaching assistant, and your instructor is the recommended approach.